
Anon from Mars, Anon from Venus: Classifying User Gender on Twitter

Yasmeena Khan and Tara Parekh
CS 129 Final Report
March 11, 2021

Abstract

Our goal is to enable gender research in the age of the Internet. Gender-focused studies are vital to understanding gender dynamics and improving equity. Since anonymity is common on the Internet, it can be difficult to conduct gender research in online environments. While past researchers have used Tweet data to classify gender (Filho et al., 2016), we decided to attempt classification based on a user's Twitter bio, in addition to their most recent Tweet, to see if we could improve accuracy over the published 63%. We used three different models to do so: logistic regression, support vector machine with Gaussian kernel, and Naive Bayes. We were able to build a classifier with 70% accuracy; a slight improvement. Our inability to improve the accuracy further points to some of the heavy caveats that encumber the task of determining the gender of anonymous users.

1 Introduction

We hope to enable gender research in online environments. To do so, we built a gender classifier for Twitter users. The input of our algorithm is the text content of a Twitter user's bio, the text content of their most recent Tweet, and their username. We compared the effectiveness of logistic regression, SVM with a Gaussian Kernel, and Naive Bayes to output a predicted gender for the user. For the purposes of this paper, we confined our gender outputs to a binary: male or female.¹

1.1 Motivation

Gender research is not a new field; many impactful studies on gender and sex have been conducted in recent years. However, the current explosion of Internet communication complicates our approach to gender research. Anonymity is a crucial part of the Internet, especially on popular websites like Twitter, Reddit, and TikTok. This makes studying gender on the Internet exceedingly difficult, and it's hard to replicate many real-world studies. How could one, for example, analyze the gender dynamics of Twitter interactions without knowing user genders? The promise of the Internet is great, especially for research purposes: there is much we stand to learn from the billions of online interactions that are publicly available. However, if we cannot glean anything about the users whose interactions we seek to analyze, our analysis can only be limited.

Therefore, our goal is to enable gender research in the age of the Internet. We will build a gender classifier that performs with above-average accuracy, so that we can successfully predict the gender of a Twitter user. We propose that a successful classifier will allow us to gather data from Twitter's API, and such data will then allow us (and future researchers) to perform more advanced gender-related research on that data. For example, in future work we might seek to analyze whether there is a difference in polarity between the tweet interactions of men and women. This relies on a solid gender classifier that can pull Twitter bio data and characterize the user's gender with moderate accuracy. In sum, our motivation is to help push forward the field of gender studies and enable it to thrive in the age of the Internet. If we solved this problem (and had a classifier that worked with full accuracy) it would enable researchers to develop ground-breaking new studies on gender in online environments.

1.2 Caveat on the Gender Binary

We have decided to characterize gender as a binary, e.g. identification as either "man" or "women," which we will use interchangeably with "male" or "female" for the purposes of this paper. In the real world, gender is not a binary; it is a

¹More details on this in Section 1.2: **Caveat on the Gender Binary.**

spectrum, with some individuals that exist outside that spectrum altogether. We do not wish to devalue or ignore this complexity in our research. The reasons we chose to classify gender as a binary are:

1. It's difficult to characterize gender as a numerical spectrum. Because there are infinite possible gender identities, it's exceedingly difficult to run a classification problem in which we adequately classify and predict people's self-identified genders;
2. Many of the real-world studies we want to replicate online use the same gender binary, so for their results it's necessary to have gender as a binary;
3. The dataset we used classified gender as a binary, as well.

You can think of gender in this project as a function of biological sex; this is not an accurate characterization of gender in the real world, but it is the simplification we had to make in order to build a useful classifier.

2 Related Work

We have already discussed several relevant papers on gender interactions in the real world, which form the basis of our motivation. Now, we turn to analyzing previous attempts at gender classification. Several studies have classified gender based on Twitter data, but few have focused on the user bio in particular. We can split the related papers into three categories based on the inputs they considered: some studies focused on individual tweets, some focused on usernames, and some focused on non-textual data types, like which colors the user used or which celebrities they followed. Across these three categories, researchers used a variety of algorithms: primarily Naive Bayes, logistic regression, and SVM.

The studies in the first category, which focused on individual tweets, are important landmarks that utilized tweet text data as n-gram features to build high accuracy classifiers (Johansson (2019)). In this paper, we are focused on classifying people rather than individual text strings. There is rich data available on a Twitter user's profile, particularly in their bios, that we are interested in utilizing. In the second category, the studies that focused on usernames (and often full names as well) were highly accurate but have limited applicability to the problem we aim to solve (Alowibdi (2013)). Once you have a user's full name, it is fairly simple to predict their gender; these studies do not solve the problem of full anonymity. Finally, the third category of studies focused on combining a wide array of non-textual data types into a classifier. For example, one paper combined five color-related features (like sidebar color and background color) to predict gender (Singh Ludu (2014)). While these studies are fascinating, we hope to build a classifier that focuses on text data. We can conclude that, while there has been meaningful work done in the Twitter gender classification space, there remains room for improvement. In particular, we hope to emphasize the importance of user bios (where a user is, by definition, describing themselves) in gender classification. We also hope to incorporate username information and the user's most recent tweet, in conjunction with this bio data, to build a stronger classifier. We identified one prior study that utilized a similarly cohesive approach to the Twitter gender problem. We consider this to be state-of-the-art: they used Naive Bayes and SVM on over 3 million training instances (Burger and Zarrella (2017)). However, their model incorporated the full name of the user, which we will not do in order to maintain our desired application. Therefore, our work contributes to the ongoing area of ML development for gender classification on Twitter, and in particular advances state-of-the-art techniques on cohesive textual user analysis.

3 Dataset

For this project, we are focusing on Twitter data, and in particular on Twitter bios in addition to usernames and Tweets.

Twitter is an American social networking and microblogging platform that allows users to post "tweets" made of up to 240 characters. Twitter has an estimated 187 million worldwide users as of 2021. Notably, Twitter allows their users to be anonymous; therefore, anonymity is common on the platform.

We chose Twitter for a few reasons:

1. It is popular among younger generations: the plurality of users on Twitter (38%) are between the ages of 18-29. This suggests that Twitter data will continue to be relevant, as opposed to data from an aging platform dominated by older users, like Facebook;
2. It has an easy-to-access API, which means we can scrape Tweet data effectively;
3. While gender on Twitter has been studied before (see: Burger and Zarrella (2017)), few studies have looked at Twitter bios (as described in the previous section) so we felt we could make a contribution in this manner.

The dataset we are working with originates from Kaggle and has 20,000 rows, each representing a different user.² Its columns are various user characteristics, including the following which are relevant to our purpose: the username of the

²Figure Eight. Twitter user gender classification, 2017. URL: <https://www.kaggle.com/crowdflower/twitter-user-gender-classification>

Twitter account, its bio, a randomly selected tweet, the gender of the account as judged by a human, and confidence in that gender.

We performed preprocessing on our data, on both the gender and textual data. The Kaggle dataset assigned each user either “male,” “female,” “brand,” or “unknown.” We began by removing all entries that were not either “male” or “female” to simplify the classification. We then changed all the “female” entries to the integer value 1, and the “male” entries to 0 to simplify the classification. The Kaggle dataset associated each entry with a “gender confidence” rating that measures how confident the human classifier was in their assignment. We chose to remove all entries with confidence ratings under 0.9 to ensure the data we operated on was gold standard. This left us with exactly 10,023 rows to train and test on.

For each of our three textual inputs, we standardized each word as lowercase and removed all punctuation. We also removed all stop words; this improved our accuracy. It was not necessary to normalize the data since all of the input features were words, and thus of the same type. We split the training and testing data 80/20, with 8,018 training instances and 2,005 testing instances. We used the `test_train_split` function in SciKit Learn in order to choose a randomly selected sample of our data to divide into the test/train data.

<i>Column Name</i>	<i>Distinct Features</i>
Screen Name	9678
User Bio	22474
User Tweet	25545
User Bio + Tweet	41157

4 Method

We used the following three models: logistic regression, SVM with a Gaussian kernel, and Naive Bayes.

4.1 Logistic Regression

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right] + \frac{\lambda}{2m} \sum_{j=1}^n \theta_j^2$$

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

Logistic regression is a popular classification algorithm. The sigmoid function turns the hypothesis (or prediction) into a value between 0 and 1, which is measured as a probability that can be turned into a class label. The cost function, provided above, measures how accurate your hypothesis was compared to the actual class. This allows you to penalize the algorithm heavily if the hypothesis is far off from the right answer. We minimize the cost function through gradient descent (repeatedly updating each parameter based on the product of the learning rate and the partial derivative of the cost function). This outputs the optimal values of θ that compose our model.

For this project, we used the “liblinear” solver in SKLearn, which minimizes the cost function by using a coordinate descent algorithm. Coordinate descent is very similar to gradient descent, but it updates one parameter at a time versus attempting to update them all at once. This choice was made for us on the defaults due to optimization time. Liblinear also includes regularization by default, which means that each parameter θ is penalized some amount in order to make sure no one parameter has outsized influence on the model.

4.2 SVM with a Gaussian kernel

$$\text{SVM hypothesis: } \min_{\theta} C \sum_{i=1}^m \left[y^{(i)} \text{cost}_1(\theta^T x^{(i)}) + (1 - y^{(i)}) \text{cost}_0(\theta^T x^{(i)}) \right] + \frac{1}{2} \sum_{j=1}^n \theta_j^2$$

$$\text{Gaussian kernel: } e^{-\frac{\|x^{(i)} - x^{(j)}\|^2}{2\sigma^2}}$$

Support vector machines are another widely used algorithm in classification problems. On a broad scale, you can think of SVMs as attempting to learn complex decision boundaries, often hyperplanes, in order to separate data that may not be linearly separable in two dimensions, but may be so in more. On a more comprehensible scale, SVMs are essentially a modified version of linear regression – see how similar their hypothesis function is to the logistic regression cost function – with a couple key differences. SVMs do not output probabilities, but rather make direct predictions of whether a given input belongs to class 0 or class 1.

The Gaussian kernel allows SVMs to learn complex nonlinear boundaries. Essentially, it uses “landmark” points to create a new feature in a dataset, where the inputs (the $x^{(i)}$ s) are modeled in relation to those landmark points; that is, a

similarity function. This can help make datasets more separable. We chose to use a Gaussian kernel because it's one of the most popular kernels for classification, because they guarantee a globally optimal predictor. Because our dataset wasn't too large, we chose to opt for this kernel, knowing it would give us the best chance at success with this model.

We used the SKLearn package, and defined the σ for the Gaussian kernel as the default, which is $\frac{1}{X.var()*n.features()}$

4.3 Naive Bayes

Bayes' Theorem as used in Naive Bayes:

$$\operatorname{argmax}_y P(Y = y | X = (x_1, x_2, \dots, x_n)) = \frac{P(X=(x_1, x_2, \dots, x_n) | Y=y) (P(Y=y))}{P(X=(x_1, x_2, \dots, x_n))}$$

The basic intuition of Naive Bayes is simple: we want to calculate which y , or class label, is most likely (that's the argmax), given each of the feature inputs. Bayes' theorem makes this problem easy; we can simply reformat it as the fraction above, and from our dataset we obtain all the probabilities necessary. The other important intuition in Naive Bayes is the assumption of independence. With so many features, this calculation would quickly become intractable, but Naive Bayes makes the assumption that all features are independent of one another. This simplification means the problems are easy to compute, and you can get an easy and powerful classifier in just a few simple steps.

We chose to use Multinomial Naive Bayes, which basically means that each variable knows its distribution is multinomial. We did this because our features were words, which means that we needed the algorithm to consider things like the frequency of certain words pertaining to a class label, which a Bernoulli Naive Bayes distribution cannot do, but at the same time, our inputs were not continuous, so a Gaussian Naive Bayes Distribution would not work, either.

We used the SKLearn Multinomial NaiveBayes package.

5 Experiments

5.1 Hyperparameters & Evaluation Metrics

Most of the hyperparameters were chosen for us through the SKLearn package; for example, it trains and tests multiple α (learning rates) and selects the optimal value. We selected an 80/20 split for the train/test set as with only $\sim 10,000$ examples, we wanted to make sure we were giving the model sufficient data to train on. We also chose to run a 5-fold cross validation as the SVM algorithm couldn't handle much more than that (even 5-fold took a few minutes to run) but we wanted to ensure we maintained high accuracy.

For our purposes, overall accuracy is the most relevant metric: the percentage of correctly classified samples. This is the metric that the previous literature on our topic has focused on. This is because we do not give preference to a particular class label; a good classifier needs to reduce errors in labeling both males and females, and hence, accuracy is the best metric. We also compute a confusion matrix to provide more insight into where our classifier performs best, and where it performs worst.

5.2 Results

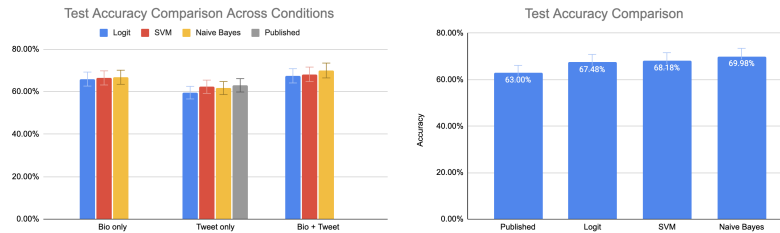


Figure 1: Accuracy Comparisons

		True gender		Total
		Female	Male	
Predicted	Female	827	255	1082
	Male	386	537	923
Total		1241	764	2005

Figure 2: Confusion matrix running Multinomial Naive Bayes on 'Bio + Tweet'

<i>Model</i>	<i>Mean Accuracy</i>	<i>Standard Deviation</i>
Logistic Regression	0.6656	0.0096
SVM	0.6721	0.0088
Naive Bayes	0.6853	0.0103

Figure 3: 5-fold cross validation over training set on 'Bio + Tweet'

5.3 Discussion

We found success by using the 'Bio + Tweet' condition, in which we considered both a user's bio and a random recent tweet as textual inputs: see Figure 1 to see the other conditions we tried, which were significantly less successful. Using that optimal condition, we were able to build a classifier under Naive Bayes with a max recorded accuracy of 69.98%, a moderately successful classifier given the limitations of determining gender over the Internet and our relatively small dataset. Notably, when we analyzed the human-level performance in our original Kaggle data, we found that the human evaluators were only fully confident in their gender predictions in approximately 13,000 out of all 20,000 examples. This is a 69% full confidence rating; notice that the success of our model is similar. This suggests that, at least in the context of our data set, there might be an upper limit on confident gender classification that is difficult to surpass.

We then ran confusion matrices (see Figure 2) on our dataset to better diagnose our errors. We found that our model over-predicted the female class, and thereby under-predicted the male class. We computed sensitivity and specificity for both classes (treating each as the "positive" case, in turn) and found that sensitivity was higher for the female class (68% versus 66%) and specificity was lower for the female class (67% versus 68%). While these values are fairly close together for both classes, they indicate that our model has a tendency to predict female too often. This might be attributable to the fact that our dataset was slightly unbalanced: 52% women to 48% men, which made it more confident in predicting the female class. It could also be related to the fact that women tend to write longer tweets. Both factors would provide the model with more data on female users to incorporate, which could increase its likelihood of predicting the female class.

Finally, we outputted probability ratings from our model to determine where it made its most confident mistakes. Where was it *sure* someone was female, and they ended up being male? (And vice versa). Combing through the tweets, we found that explicitly gendered language used by the opposite sex tended to confuse our model. For example, when a female user was describing someone of the opposite gender and used the words "he" and "his," our model often predicted the user as being male. On the other hand, we saw many men with bios that included lines like "lover of women," which our model incorrectly classified as female. Our model was also confused by gendered words used out of context, and in particular, religion: when female users had words like "Him," "His," or male names like "Muhammad" in their bios (often in a religious context) our model often predicted them as male. Finally, our model tended to misclassify shorter tweets and bios more often, likely because there was less information to analyze.

The main takeaways are that the dataset was slightly unbalanced and that our model had some pitfalls related to gendered terminology. We also wanted to make sure that we weren't overfitting to our training set, so we ran 5-fold cross validation; the results showed the mean accuracy was extremely close, and in fact all of our outputs were within a standard deviation of the mean accuracy.

6 Conclusion and Future Work

In conclusion, we achieved moderate success with our model, constructing an algorithm that performed significantly above chance (nearly 70% accuracy), and moreover outperformed some of the earlier research in the field. Naive Bayes performed the best out of the algorithms we tried. We have a few ideas about why. For one, Naive Bayes' simplicity assumption may have helped in this exercise, as assuming words are independent of each other allows words that are used more by one gender become more relevant in a model. Secondly, the SVM might have worked better with a different σ or kernel, but we were unable to utilize it further as it was too inefficient to handle large amounts of data. Finally, all the algorithms were actually quite close in accuracy, so some of the difference may have simply been noise.

However, 70% accuracy is not nearly good enough for professional research studies on gender. For future work, we'd want to include a confidence metric in our classifier that would report how confident we are in classifying a given user. This would allow us to only classify users with >90% confidence, which we hypothesize would lead to much higher accuracy. We'd also like to use the Twitter API to pull more tweets and see if using more features helps the accuracy of the model; we didn't necessarily think it would, but it's possible that it might. Finally, as we continue to develop our classifier, we'd love to apply it to real gender studies in online environments.

References

- Fredrik. Johansson. Supervised classification of twitter accounts based on textual content of tweets. *CLEF*, 2019.
- J. S. Alowibdi. Empirical evaluation of profile characteristics for gender classification on twitter. *International Conference on Machine Learning and Applications*, 2013.
- Puneet Singh Ludu. Inferring gender of a twitter user using celebrities it follows. *Cornell CSE*, 2014.
- John Burger and Guido Zarrella. Discriminating gender on twitter. *The MITRE Corporation*, 2017. URL http://www.mitre.org/sites/default/files/pdf/11_0170.pdf.
- Bruce Goldman. Two minds: The cognitive differences between men and women, 2017. URL <https://stanmed.stanford.edu/2017spring/how-mens-and-womens-brains-are-different.html>.
- Dean H. Zimmerman and Candace West. Sex roles, interruptions and silences in conversation. *Towards a Critical Sociolinguistics*, 125, 1996. doi: 10.1075/cilt.125.12zim.
- Tonja Jacobi and Dylan Schweers. Justice, interrupted: The effect of gender, ideology and seniority at supreme court oral arguments. *Virginia Law Review*, 1379, 2017. doi: <https://ssrn.com/abstract=2933016>.
- Figure Eight. Twitter user gender classification, 2017. URL <https://www.kaggle.com/crowdflower/twitter-user-gender-classification>.